UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
Department of Physics and Astronomy

Electron Diffraction

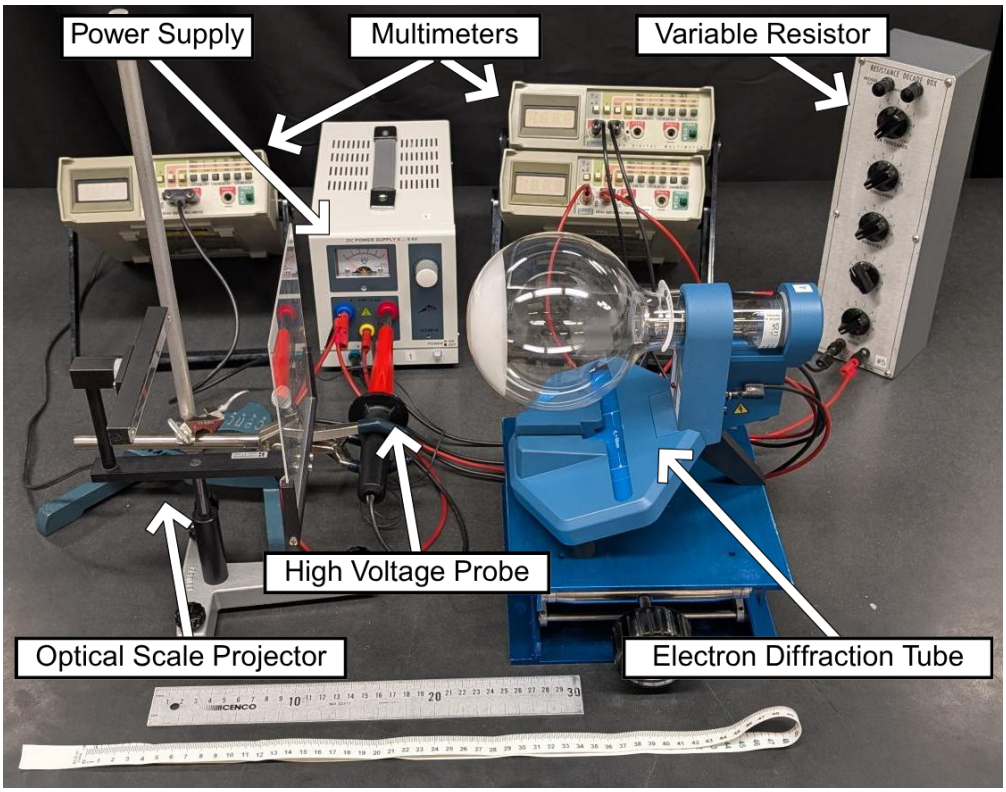


Figure 1.1. Electron diffraction experiment setup.

Equipment: Fluke multimeter [3], high voltage power supply, 100 Ω GR decade resistance box, high voltage probe, illuminated optical scale projector, electron diffraction tube, laboratory jack, He-Ne laser, laboratory stand [2], right angle clamp [3], fork clamp [2], electron diffraction optical analogue wheel, set of connecting leads, imperial threaded aluminum mounting rods.

Goals of the Experiment: To demonstrate that electrons interact with a crystal in a manner consistent with waves rather than classical particles and produce a diffraction pattern; To determine the lattice constants of carbon.

Theory

A beam of light is a wave, but it can exchange energy and momentum with matter at points via photons. In 1924, Louis de Broglie suggested that a beam of moving particles can be thought of as a matter wave that transfers energy and momentum to other matter at points. In 1927 this hypothesis was proven experimentally when a diffraction pattern was observed from the electrons scattered by crystalline nickel.

The length λ of a wave associated with a moving electron of momentum p is

$$\lambda = \frac{h}{p} = \frac{h}{m_e v}, \tag{1.1}$$

where h is Planck's constant ($h = 6.626 \times 10^{-34}$ J $\cdot$ s) , $m_e = 9.11 \times 10^{-31}$ kg is the mass of an electron and v is the speed of the electron.

The energy E of the electron can be expressed as

$$E = \frac{p^2}{2m_e} = hf, \tag{1.2}$$

where f is the frequency of the wave associated with the electron.

An electron of charge $e = 1.602 \times 10^{-19}$ C accelerated by a voltage V_a has the energy

$$E = eV_a. \tag{1.3}$$

Combining equations 1.2 and 1.3 allows us to express the momentum of the particle as

$$p = \sqrt{2m_e eV_a}. \tag{1.4}$$

The de Broglie wave associated with the electron can be then written as:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV_a}} \tag{1.5}$$

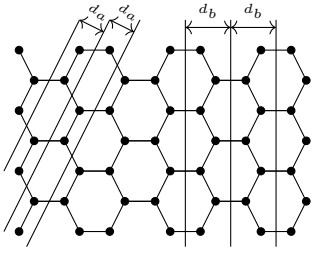


Figure 1.2. Carbon Crystalline Structure

In this experiment, electrons, emerging from the cathode due to thermionic emission, are subjected to an accelerating potential V_a and are incident on a carbon film target that acts like a diffraction grating for the electrons. The carbon film target consists of randomly oriented sheets of carbon atoms. In each sheet, the atoms are in a hexagonal arrangement which provides two distinct diffraction planes, as shown in Figure 1.2, with the spacings $d_a = 0.123$ nm and $d_b = 0.213$ nm. The random orientation of the carbon atom sheets corresponds to the rotating of the optical diffraction grating, hence resulting circular pattern will be produced. As the carbon crystal effectively creates two diffraction gratings, two separate series of maxima are anticipated in the resulting diffraction pattern. Due to the physical limitations of the setup, only the first order maxima will be possible to observe.

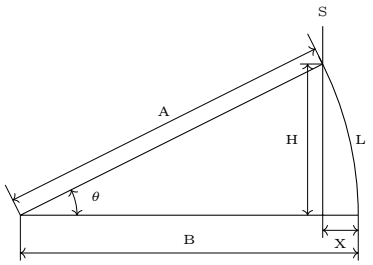


Figure 1.3. Path Length Difference, A: Diffraction path length, B: Target screen distance, H: Ring radius, L: Luminescent screen, S: Diffraction plane, θ: Diffraction angle

To observe a diffraction pattern on screen S in Figure 1.3 it is necessary that

$$\lambda = d\sin\theta, \quad \text{(n=1 for first order maxima)}, \tag{1.6}$$

which, for the triangle in Figure 1.3, can be written as

$$\lambda = d\sin\theta = \frac{d \cdot H}{A} \tag{1.7}$$

where d is the lattice spacing constant, H is the diffraction ring radius, A is the diffraction path length. From the geometry of the setup, one can deduce that the length A can be expressed as

$$A = \sqrt{H^2 + (B - X)^2}, \tag{1.8}$$

where B is the target-screen distance and X is the diffraction plane to screen distance calculated using:

$$X = \text{Radius} - |CD| = R - \sqrt{R^2 - H^2}. \tag{1.9}$$

Combining equations 1.5 and 1.7 allows to relate the diffraction radius H and the accelerating potential V_a as follows:

$$H = \frac{Ah}{d\sqrt{2m_e eV_a}}. \tag{1.10}$$

Equation 1.10 can be used to calculate the lattice spacing constants d_a and d_b for a carbon lattice.

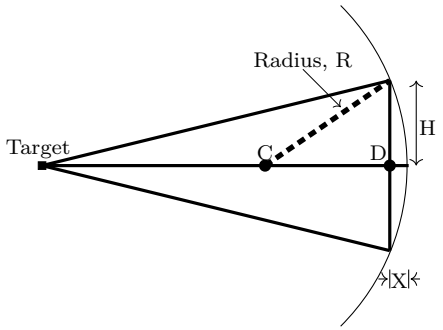


Figure 1.4. Bulb Geometry, C: Bulb Centre, D: Diffraction Plane, H: Ring Radius, X: Diffraction plane to Luminescent screen distance

The Experiment

Uncertainty Estimation and Propagation

Uncertainties in H originate from simple measurements and can be estimated individually. Uncertainties is A and X have to be determined through uncertainty propagation. Uncertainties for voltage measurements, however, result from instrument errors for both the high voltage probe and the Fluke multimeter, and must be considered jointly. See the back panel on the multimeter. The probe has an uncertainty of 1% of your total reading. Propagate uncertainties using appropriate methods.

Pre-Lab

Your Pre-lab exercises are available on the D2L Lab site (Assessment – Quizzes) and should be completed there.

Experimental Procedure

$\Longrightarrow$ **Step 1** A demonstration of a two dimensional grating will be shown by your TA during the introduction to the lab. When illuminated with a He-Ne laser, this grating will produce a 2D array of dots on the screen/wall. The grating is free to rotate, once it is set spinning the rectangular array of dots appears to merge into concentric circles. This demonstrates what we expect to see on the luminescent screen by diffracted electrons. The structure of crystal lattice in the electron diffraction tube is imperfect (i.e. small sections are at random orientations to one another) which produces continuous circles instead of discrete bright spots.

$\Longrightarrow$ **Step 2** Using the soft tape measure, measure the circumference and infer the radius of the bulb R with uncertainty. Assume that the target-screen distance B is the same as the diameter of the bulb.

$\Longrightarrow$ **Step 3** The circuit shown in Figure 1.5 has been set up for you, but should be verified before taking measurements. As a group, compare the circuit diagram to the circuit on the bench and then confirm with your TA before applying power. Then, set the variable resistance to 20 Ω and ensure the control knob on the DC power supply is initially set to zero (turned fully

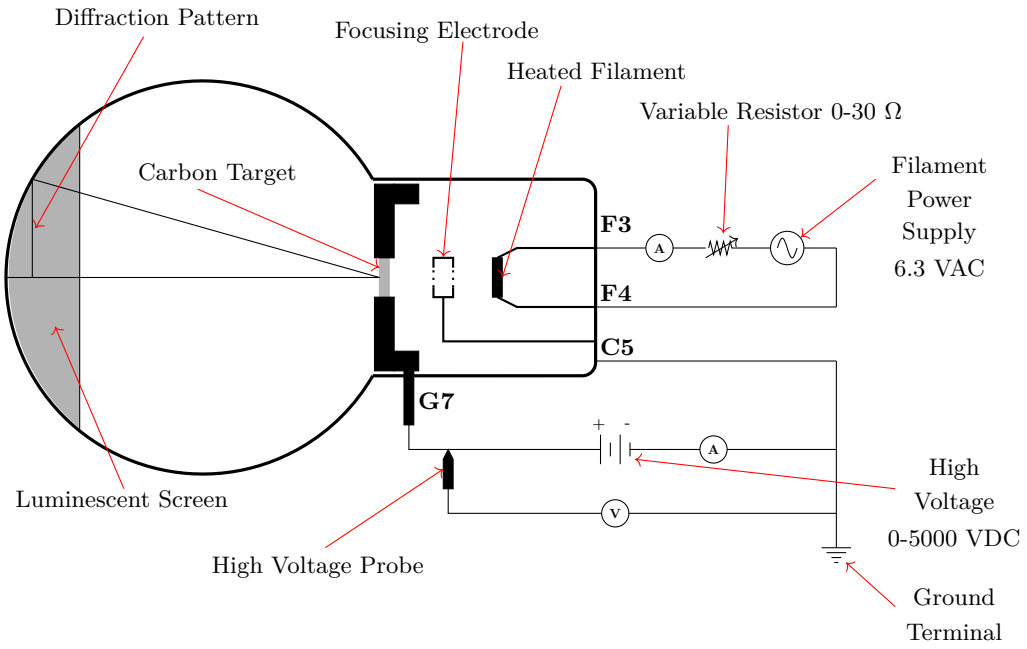


Figure 1.5. Experimental Setup

counter-clockwise), then turn it on. This will start the 6.3 VAC heater power supply (see Figure 1.5 for reference).

⟹ **Step 4** The ammeter for the high voltage is used to monitor the emission current. If this current excedes 0.3 mA it can destroy the graphite foil on the diffraction grating. Turn off the power supply and contact your TA if this occurs.

⟹ **Step 5** Be extremely careful with any high voltage power supplies. The high voltage probe needs to be fully seated in the positive terminal of the power supply; if it is pushed out slightly the voltage will arc within and make a buzzing sound, push it back in using the handle if needed. Slowly increase the accelerating voltage V_a to about 5.0 kV (voltmeter will display it in kV). At this voltage, the intensity of the screen pattern will be at its maximum. It may be necessary to adjust the heater current (by adjusting the variable resistor) in order to obtain easy legibility of the scale divisions on the optical scale projector (increase resistance to reduce brightness). The heater has a thermal response time of about 30 seconds. Use the adjustable magnet to focus the electron beam at the centre of the tube face before taking any measurements.

⟹ **Step 6** Due to the small size of the diffraction tube, a central dot and two rings will be visible. Both rings are of the first order ($n = 1$) and correspond to the two grating constants d_a and d_b of the carbon film. Determine which ring corresponds to which grating constant (d_a or d_b) using Equation 1.10. The diffraction rings are observed on the inner spherical surface of the diffraction tube. The carbon target is not positioned at the centre of tube curvature and, as a consequence, A must be calculated for each value of H separately.

⟹ **Step 7** Measure H for both rings using the supplied optical scale projector. The illuminated scale produces a virtual image extending out into space beyond the plexiglass end of the instrument. The virtual image can be positioned in the plane of the diffraction rings for very accurate readings. Align the scale projector perpendicular to the axis to measure H ensuring that the

plexiglass is halfway between the illuminated scale and the bulb. Take $8-10$ measurements of V_a from 5.0 kV and decrementing by $0.2-0.25$ kV (see Excel template).

Data Requirements

- A table of values of X, H, V_a, and A with uncertainties. Use the SI system of units and specify uncertainties.
- Graphs of $A/\sqrt{V_a}$ versus H **for each diffraction ring**.

Lab Report

Your lab report should include the following items:

- A table including measurements and calculations of all distances listed in Equations 1.7 and 1.9 with uncertainties.
- Rearrangements of Equation 1.10 to show dependence of $A/\sqrt{V_a}$ on H.
- Plots of $A/\sqrt{V_a}$ versus H for each of the two diffraction rings.
- Results of the linear fit of each of the graphs with their uncertainties.
- Calculations of lattice constants d_a and d_b and their uncertainties.
- Comparison of the lattice constants d_a and d_b with accepted values (see the introduction).